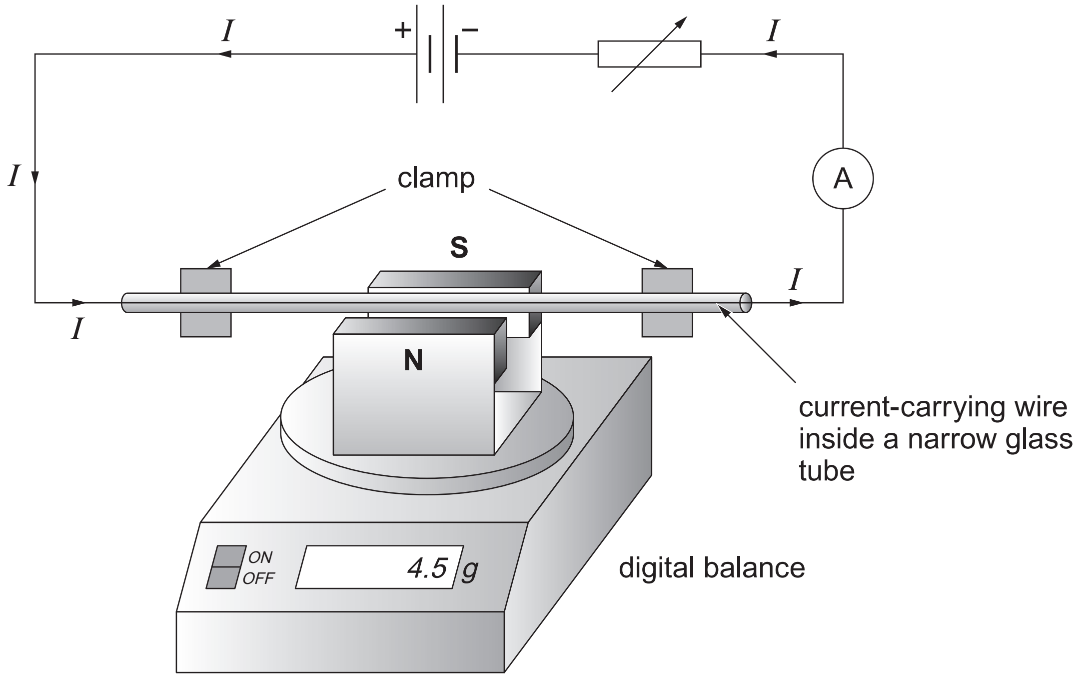
Examiner only

7. Apparatus is set up in a laboratory to investigate the force exerted by a current-carrying wire when it is placed in a uniform magnetic field. A magnet is placed on a digital balance and a current-carrying wire is placed between its magnetic poles.



(a) To prevent the current-carrying wire moving during the experiment it is contained in a narrow glass tube that is clamped. A pupil predicts that the current-carrying wire would move upwards if it had **not** been contained in the narrow glass tube.
Explain whether the pupil is correct. [3]

....

....

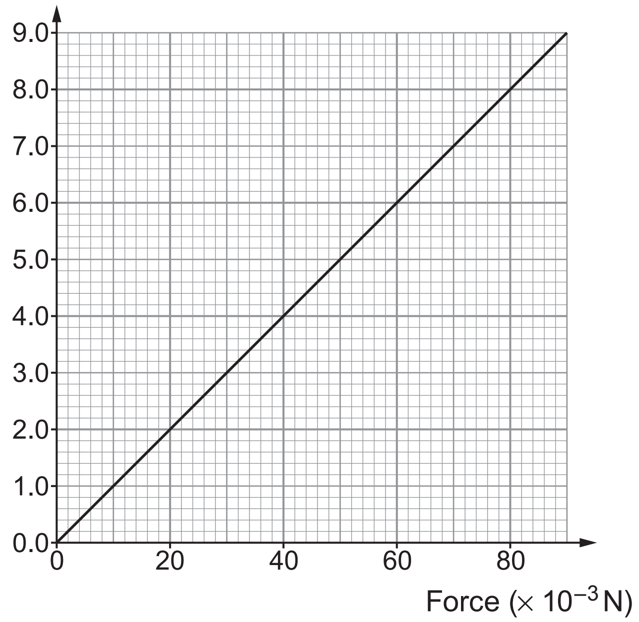
....

....

....

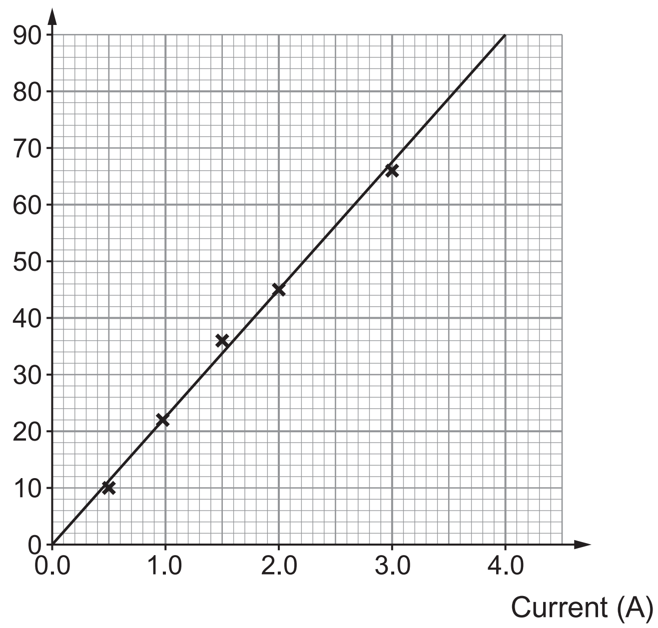
(b) When **no** current flows through the wire the digital balance is adjusted to read zero. The mass reading displayed on the digital balance is converted to a force, in newtons, using **Graph 1**.

Graph 1



The current through the wire is varied and the readings of force calculated. The data is plotted on the grid below and a best fit straight line is drawn.

Graph 2



Examiner only

(i) During the experiment a reading of **4.0 g** is displayed on the digital balance. Use the graphs opposite to complete the table. [2]

Mass (g)	Force (N)	Current (A)
4.0	....	....

(ii) Describe the relationship between current and force. [2]

....

....

....

(iii) A student correctly states that the line on **Graph 2** must obey the equation of a straight line: $y = mx + c$. Comparing this with $F = BIl$ and, given that force is plotted on the y-axis, identify the **quantities** that represent the gradient and the intercept. [2]

Gradient = ...

Intercept = ...

(iv) The length of wire contained in the magnetic field, B, is 5.0 cm. Use information from **Graph 2** to calculate the magnetic field strength of the magnet used in the experiment. [3]

Magnetic field strength, B = ... T

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